

Overview of Antibiotic Classes and Individual Antibiotics

Twisha Patel, PharmD, BCPS
Clinical Specialist, Infectious Diseases
Adjunct Clinical Instructor
University of Michigan Health System
University of Michigan College of Pharmacy



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Various Mechanisms of Action

Cell Wall Synthesis

Beta Lactams

Penicillins
Cephalosporins
Carbapenems
Monobactams

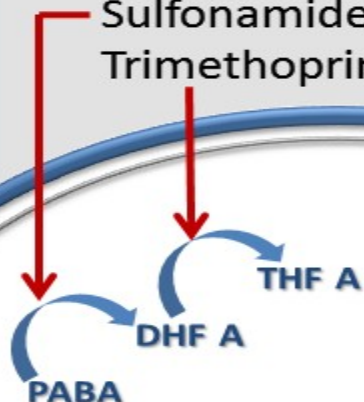
Vancomycin
Bacitracin

Cell Membrane

Polymyxins

Folate synthesis

Sulfonamides
Trimethoprim



Nucleic Acid Synthesis

DNA Gyrase

Quinolones

RNA Polymerase

Rifampin



50S

30S

50S subunit

Macrolides
Clindamycin
Linezolid
Chloramphenicol
Streptogramins

30S subunit

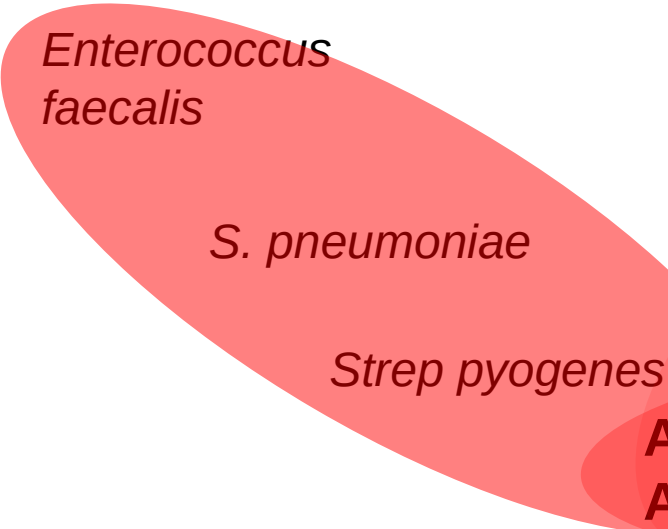
Tetracyclines
Aminoglycosides

Protein Synthesis

Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)



*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Serratia

*Klebsiella
pneumoniae*

E. coli

*Proteus
mirabilis*

H. influenzae

Enterobacter spp

*Bacteroides
fragilis*

Pseudomonas

Ampicillin
Amoxicillin

Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Nafcillin

*Proteus
mirabilis*

E. coli

*Klebsiella
pneumoniae*

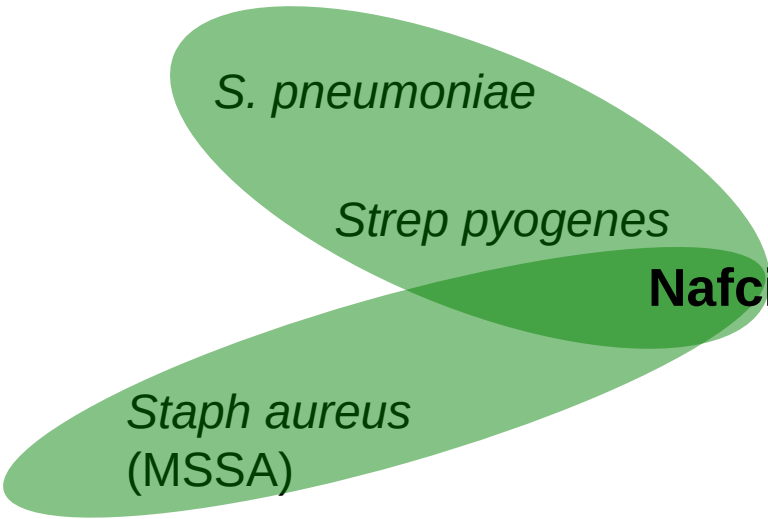
Serratia

H. influenzae

Enterobacter spp

*Bacteroides
fragilis*

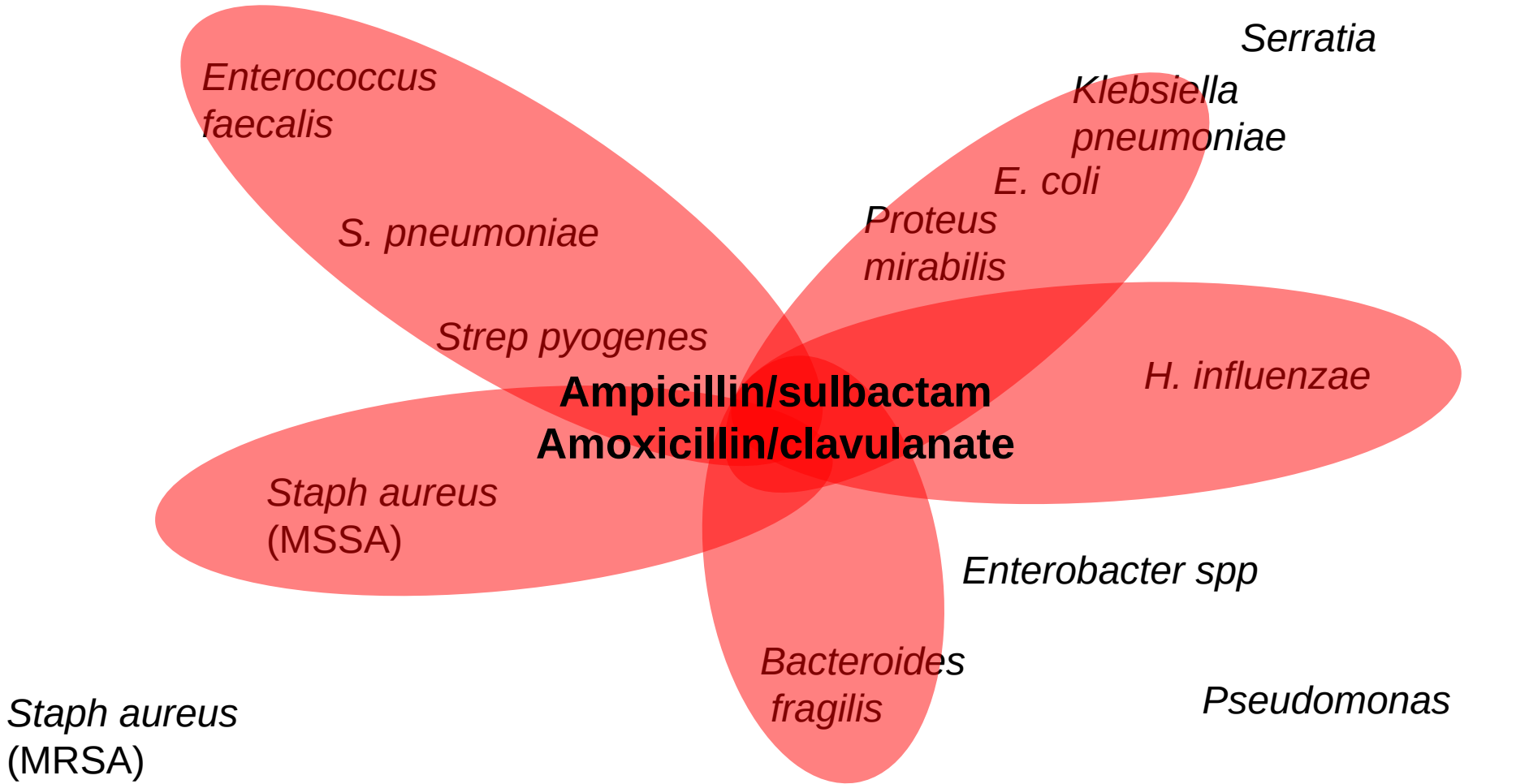
Pseudomonas



Gram-positive

Gram-negative

Enterococcus faecium (VRE)

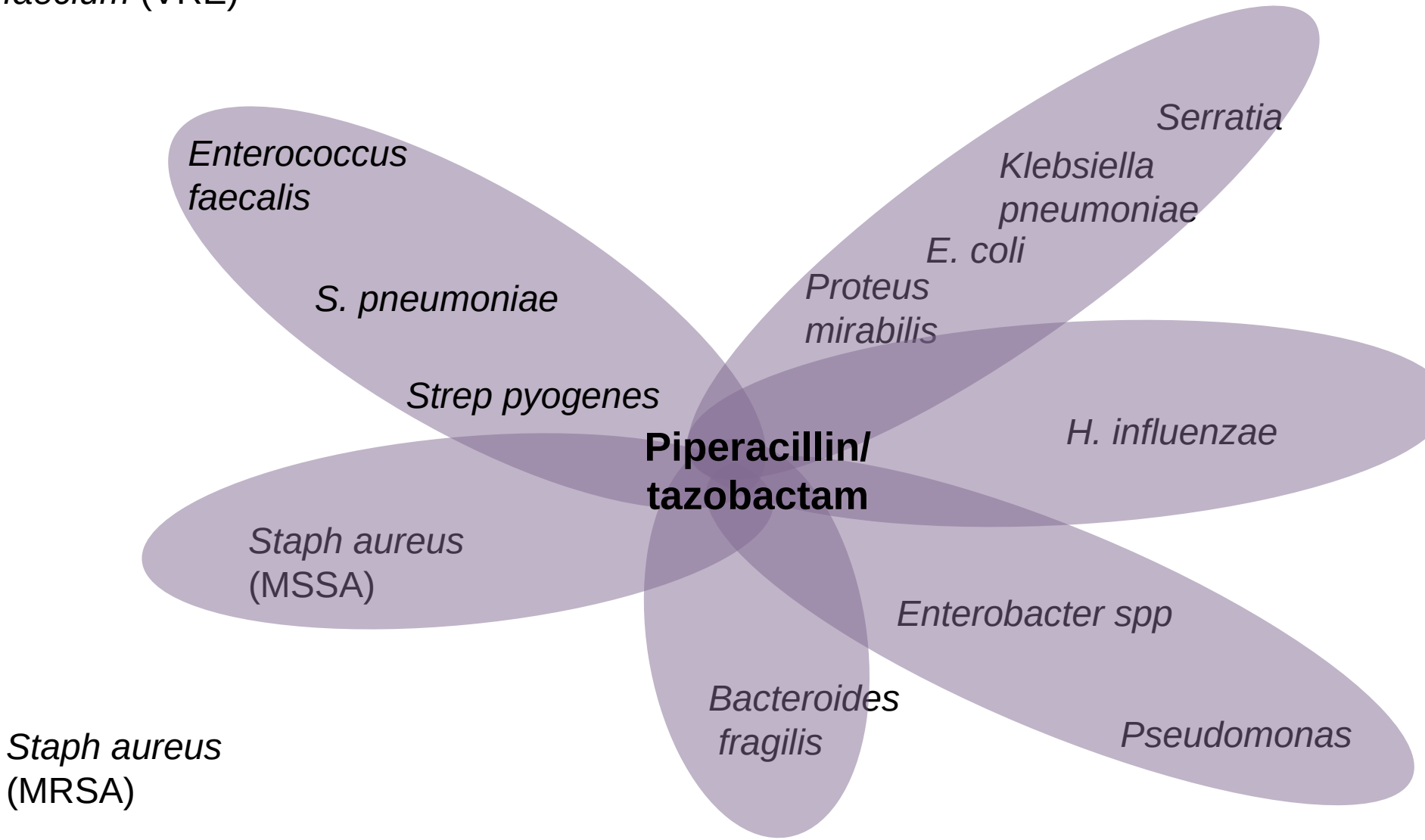


Staph aureus (MRSA)

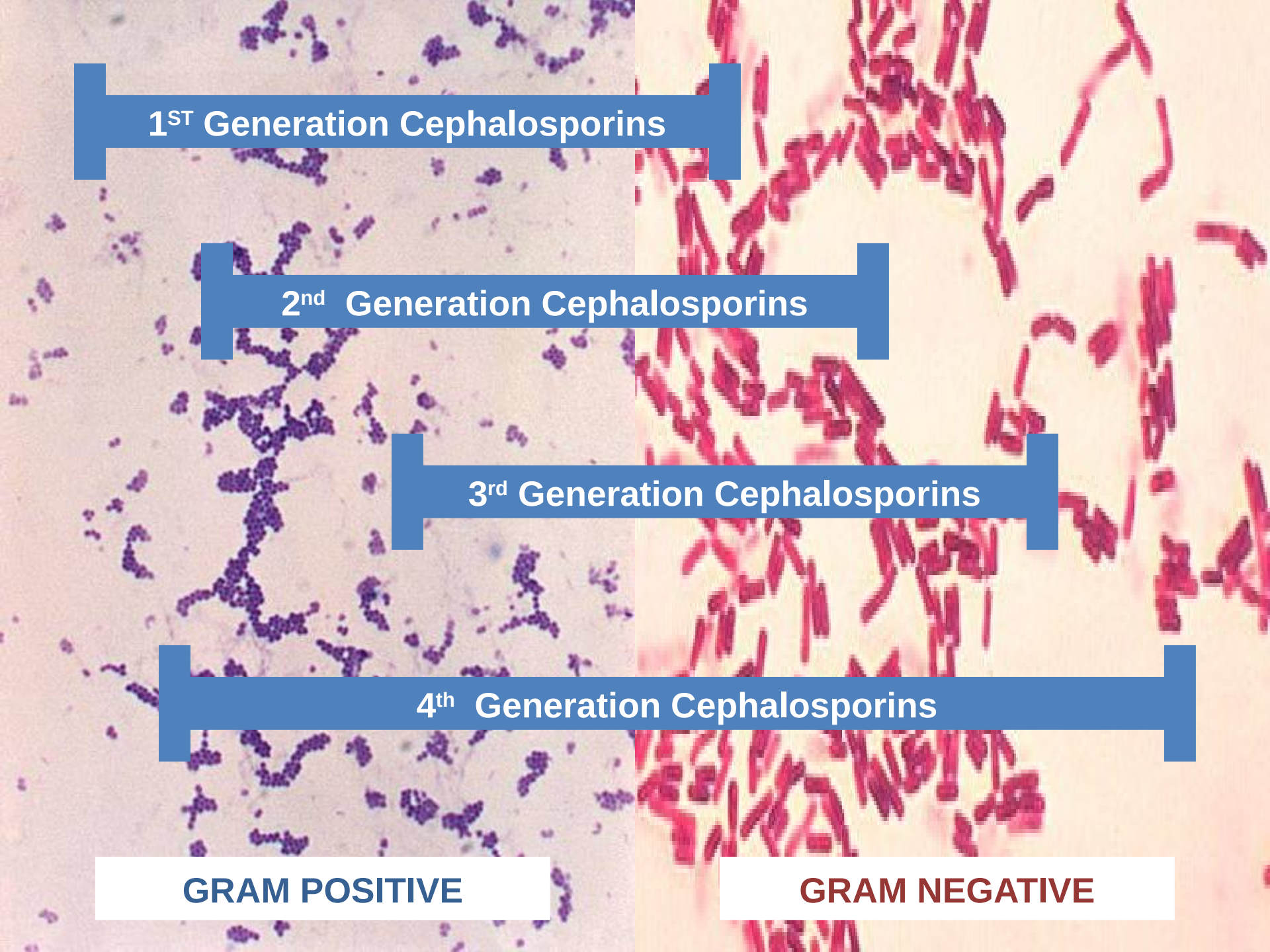
Gram-positive

Enterococcus faecium (VRE)

Gram-negative



Staph aureus (MRSA)

A microscopic image showing Gram-stained bacteria. The left side of the image contains purple-stained Gram-positive bacteria, appearing as small, irregular clusters. The right side contains pink-stained Gram-negative bacteria, appearing as long, thin, rod-shaped structures. Four blue horizontal bars with white text are overlaid on the image, spanning from the purple to the pink area. The text on the bars indicates the generations of cephalosporins: 1st, 2nd, 3rd, and 4th. At the bottom, two white boxes with blue and red text identify the Gram-positive and Gram-negative regions.

1ST Generation Cephalosporins

2nd Generation Cephalosporins

3rd Generation Cephalosporins

4th Generation Cephalosporins

GRAM POSITIVE

GRAM NEGATIVE

Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Cefazolin

*Bacteroides
fragilis*

*Proteus
mirabilis*

E. coli

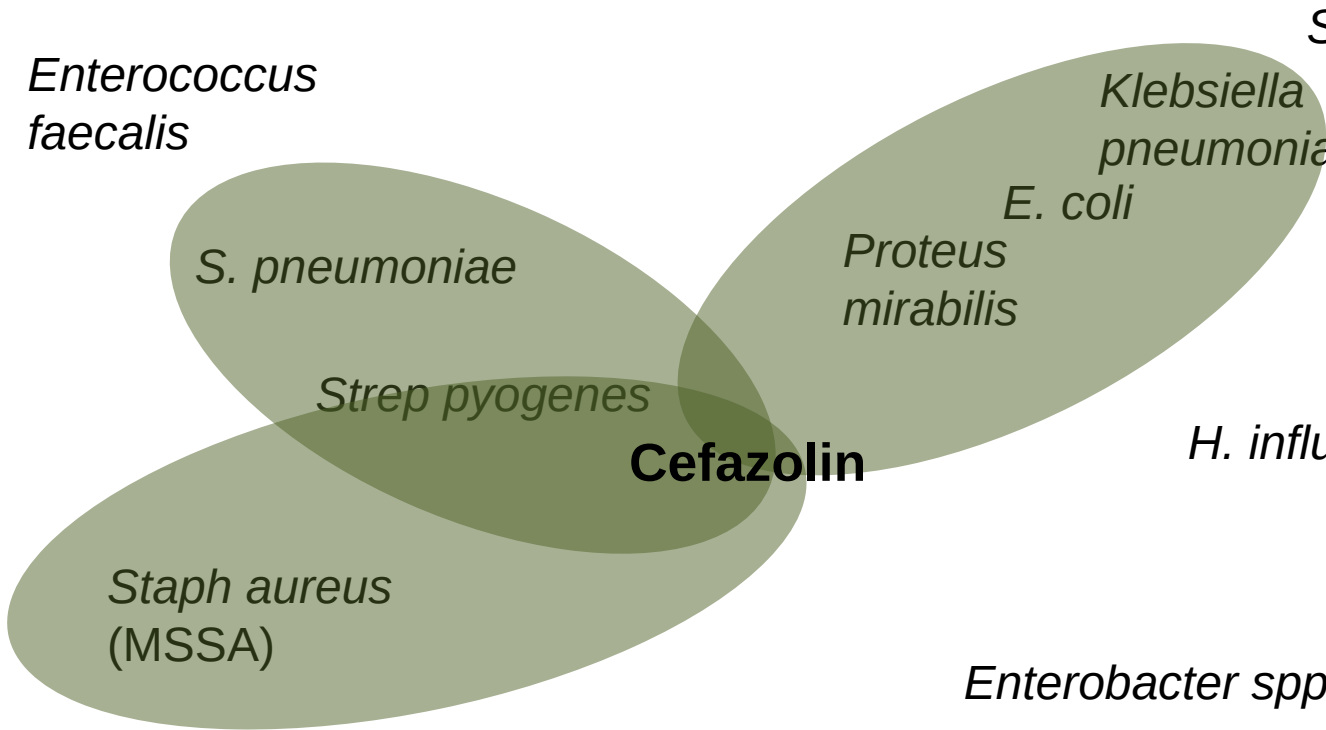
*Klebsiella
pneumoniae*

Serratia

H. influenzae

Enterobacter spp

Pseudomonas



Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Cefuroxime

*Bacteroides
fragilis*

*Proteus
mirabilis*

E. coli

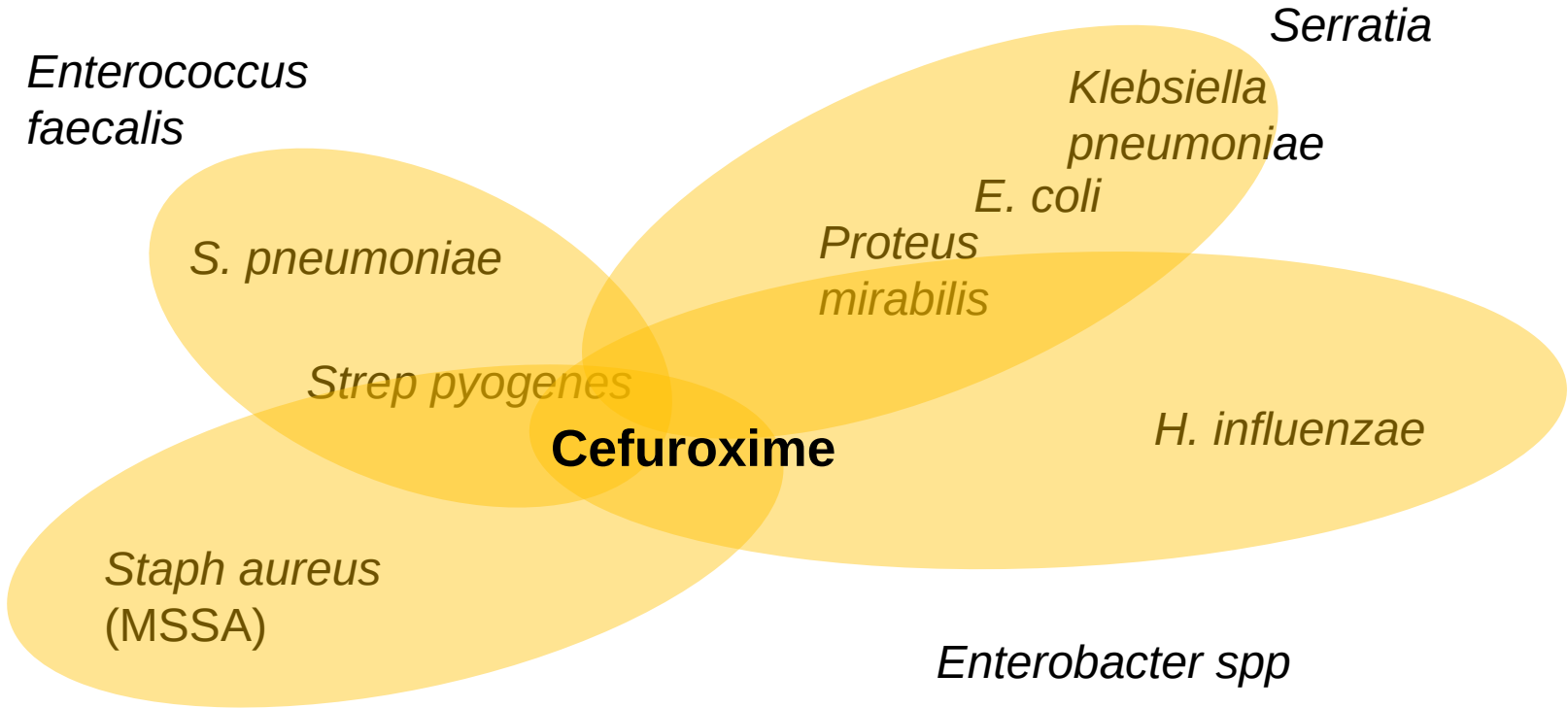
*Klebsiella
pneumoniae*

Serratia

H. influenzae

Enterobacter spp

Pseudomonas



Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Ceftriaxone
Cefotaxime

*Proteus
mirabilis*

E. coli

*Klebsiella
pneumoniae*

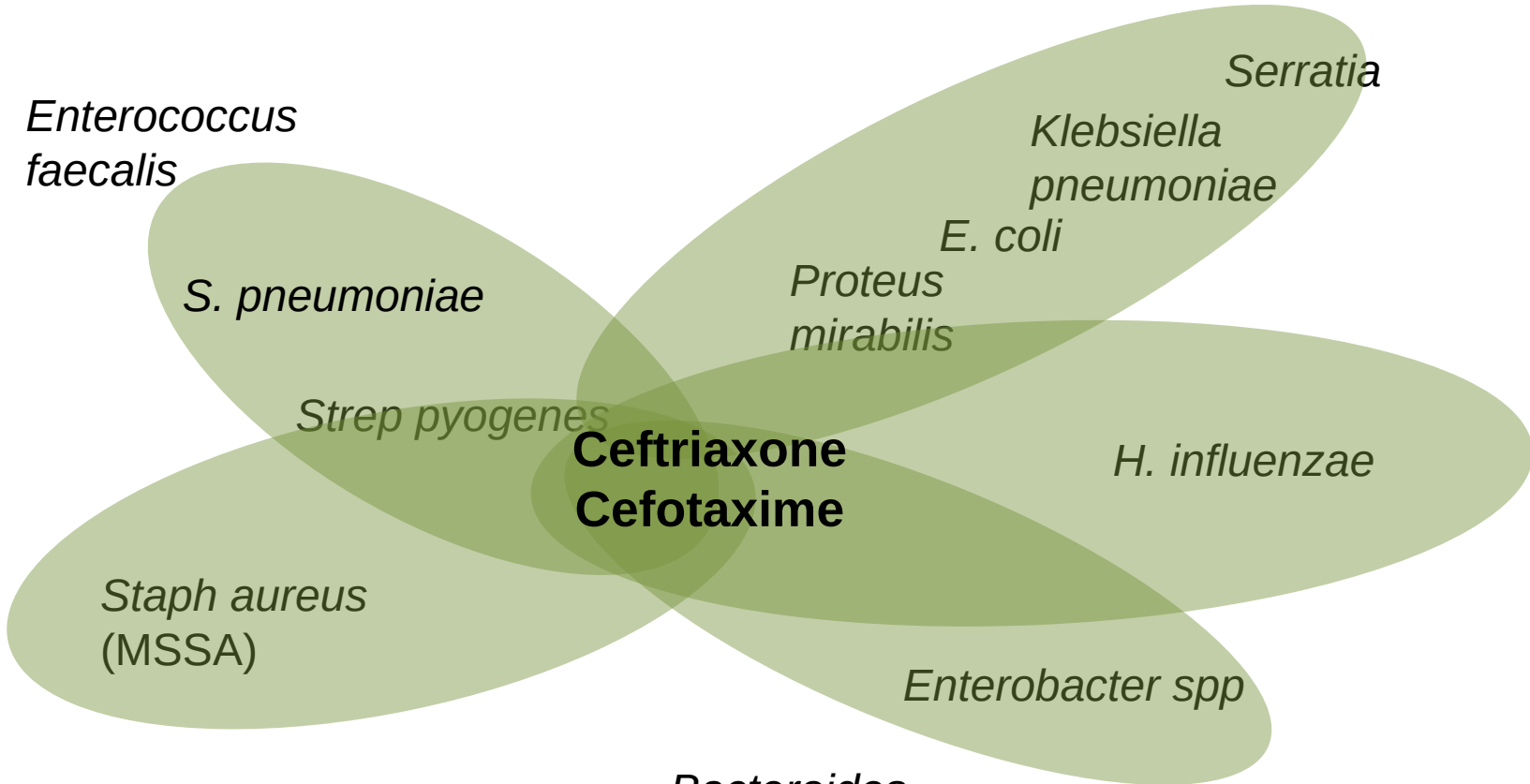
Serratia

H. influenzae

Enterobacter spp

*Bacteroides
fragilis*

Pseudomonas



Gram-positive

Enterococcus faecium (VRE)

Enterococcus faecalis

S. pneumoniae

Strep pyogenes

Staph aureus (MSSA)

Staph aureus (MRSA)

Gram-negative

Serratia

Klebsiella pneumoniae

E. coli

Proteus mirabilis

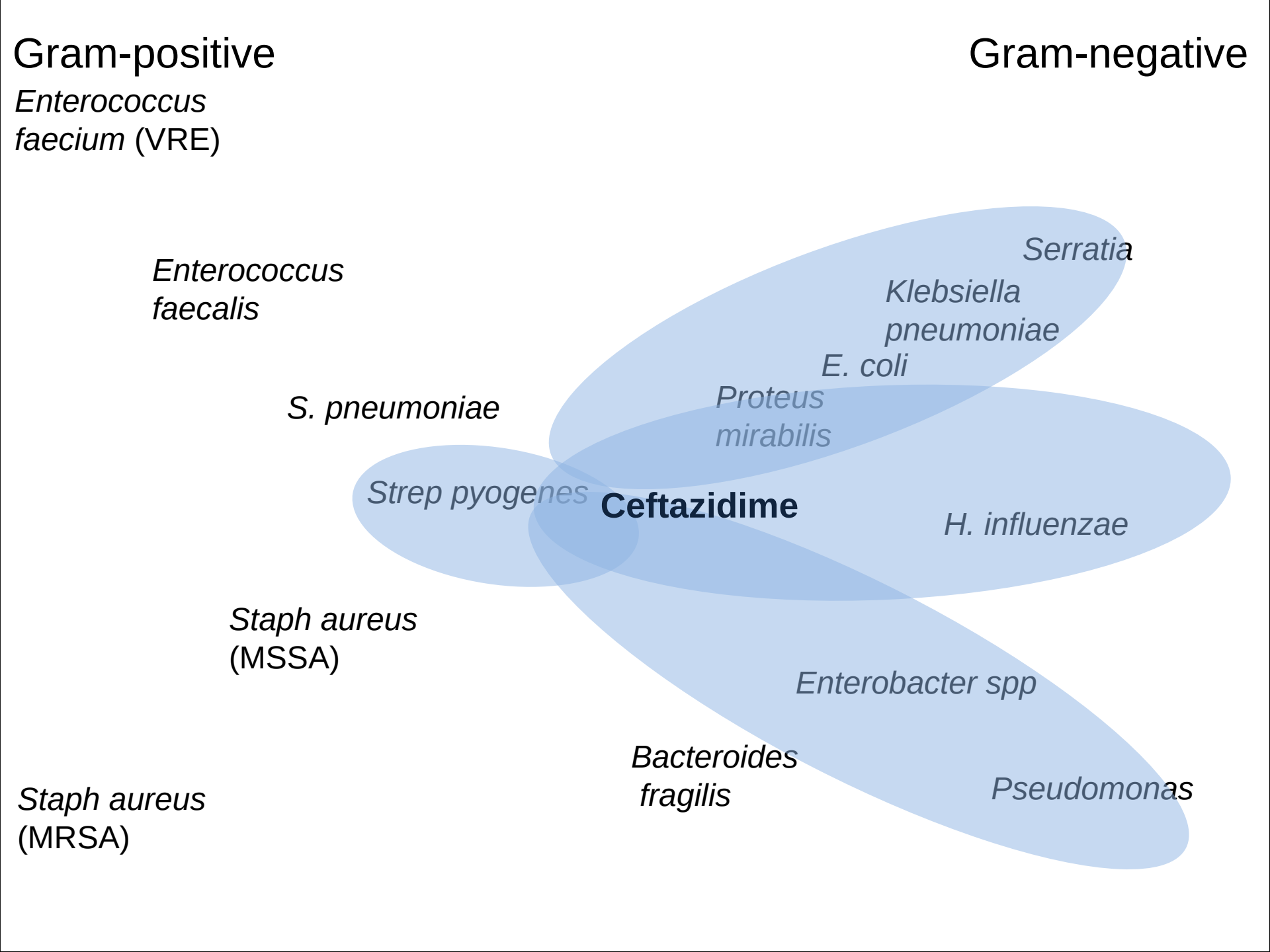
H. influenzae

Enterobacter spp

Bacteroides fragilis

Pseudomonas

Ceftazidime



Gram-positive

Gram-negative

Enterococcus faecium (VRE)

Enterococcus faecalis

S. pneumoniae

Strep pyogenes

Staph aureus (MSSA)

Staph aureus (MRSA)

Cefepime

Proteus mirabilis

E. coli

Klebsiella pneumoniae

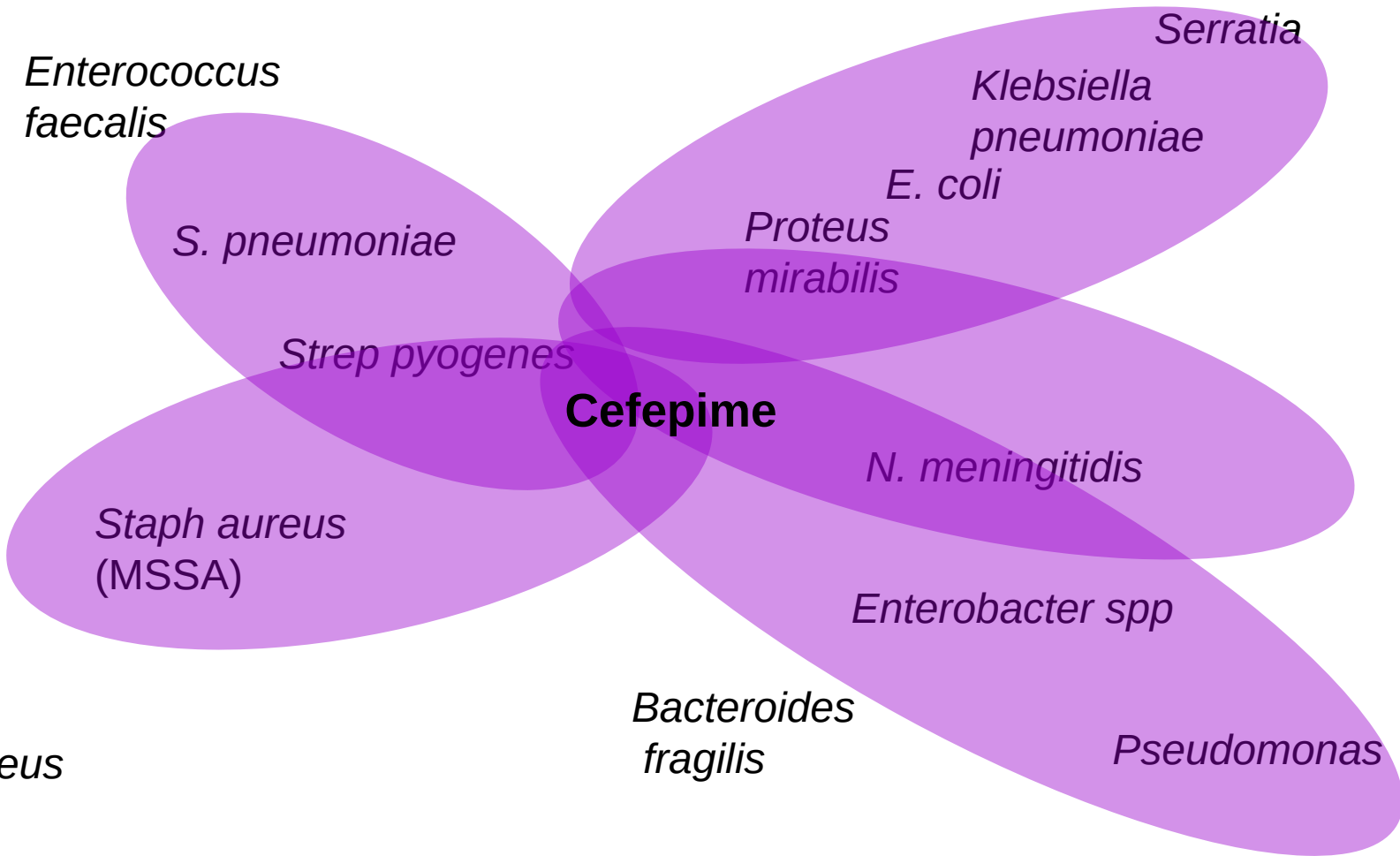
Serratia

N. meningitidis

Enterobacter spp

Bacteroides fragilis

Pseudomonas



Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Ertapenem

*Proteus
mirabilis*

E. coli

*Klebsiella
pneumoniae*

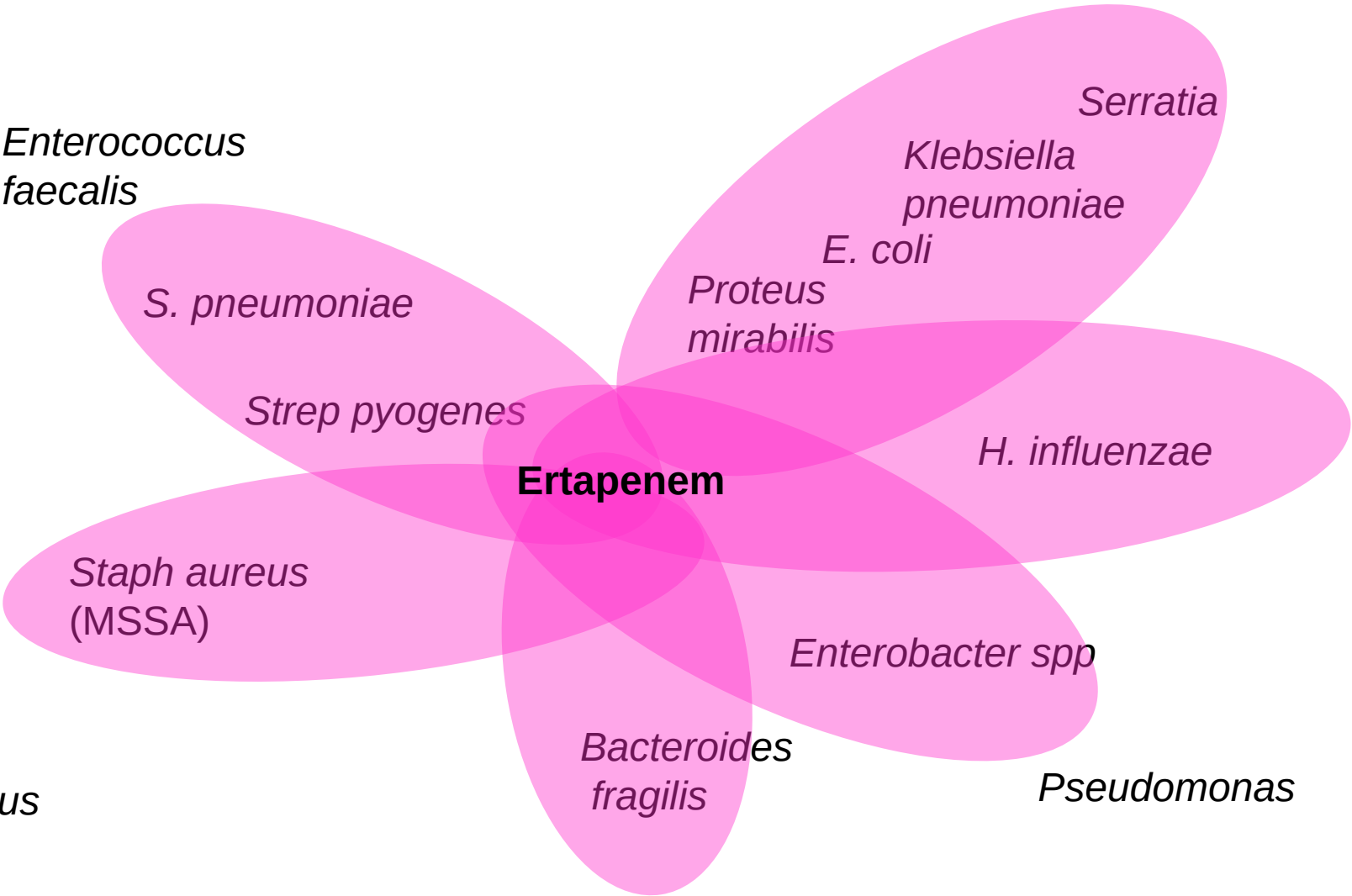
Serratia

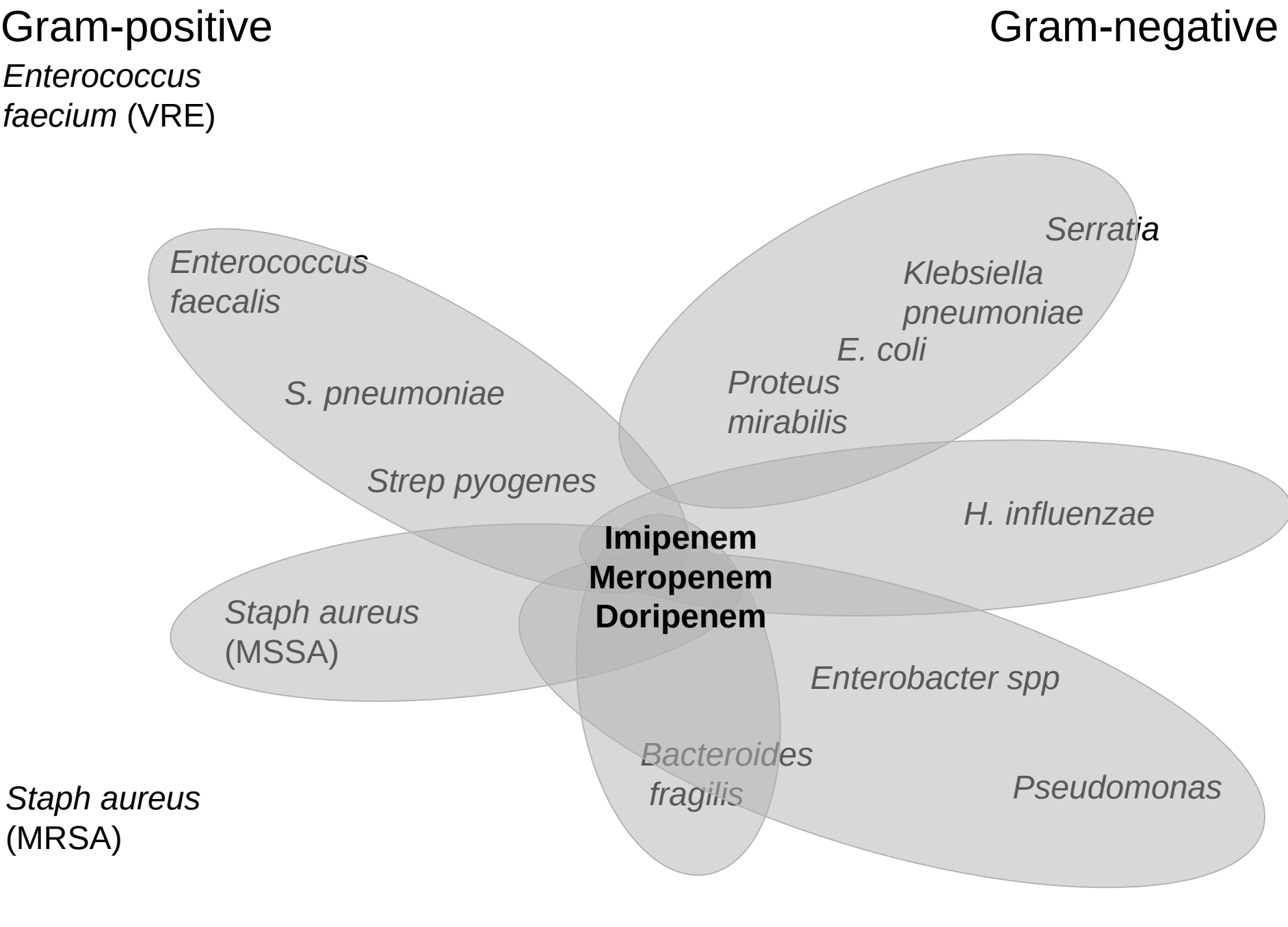
H. influenzae

Enterobacter spp

*Bacteroides
fragilis*

Pseudomonas





Gram-positive

Enterococcus faecium (VRE)

Enterococcus faecalis

S. pneumoniae

Strep pyogenes

Staph aureus (MSSA)

Staph aureus (MRSA)

Gram-negative

Serratia

Klebsiella pneumoniae

E. coli

Proteus mirabilis

H. influenzae

N. meningitidis

Enterobacter spp

Bacteroides fragilis

Pseudomonas

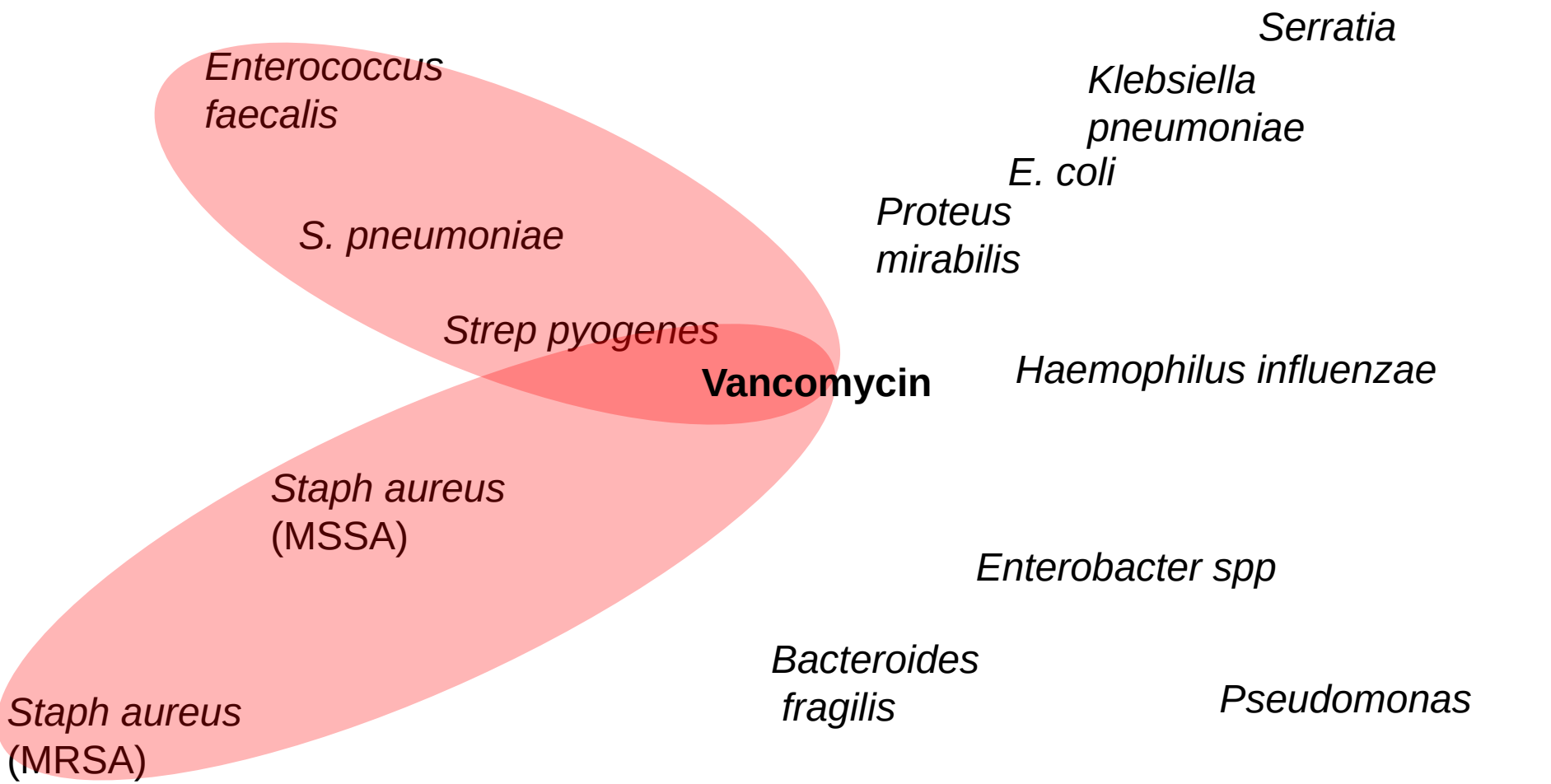
Aztreonam

Gram Type	Bacterium	Aztreonam Activity
Gram-positive	<i>Enterococcus faecium</i> (VRE)	No
Gram-positive	<i>Enterococcus faecalis</i>	No
Gram-positive	<i>S. pneumoniae</i>	No
Gram-positive	<i>Strep pyogenes</i>	No
Gram-positive	<i>Staph aureus</i> (MSSA)	No
Gram-positive	<i>Staph aureus</i> (MRSA)	No
Gram-negative	<i>Serratia</i>	Yes
Gram-negative	<i>Klebsiella pneumoniae</i>	Yes
Gram-negative	<i>E. coli</i>	Yes
Gram-negative	<i>Proteus mirabilis</i>	Yes
Gram-negative	<i>H. influenzae</i>	Yes
Gram-negative	<i>N. meningitidis</i>	Yes
Gram-negative	<i>Enterobacter spp</i>	Yes
Gram-negative	<i>Bacteroides fragilis</i>	No
Gram-negative	<i>Pseudomonas</i>	Yes

Gram-positive

Gram-negative

Enterococcus faecium (VRE)



Vancomycin

Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Daptomycin

*Bacteroides
fragilis*

*Proteus
mirabilis*

E. coli

*Klebsiella
pneumoniae*

Serratia

Haemophilus influenzae

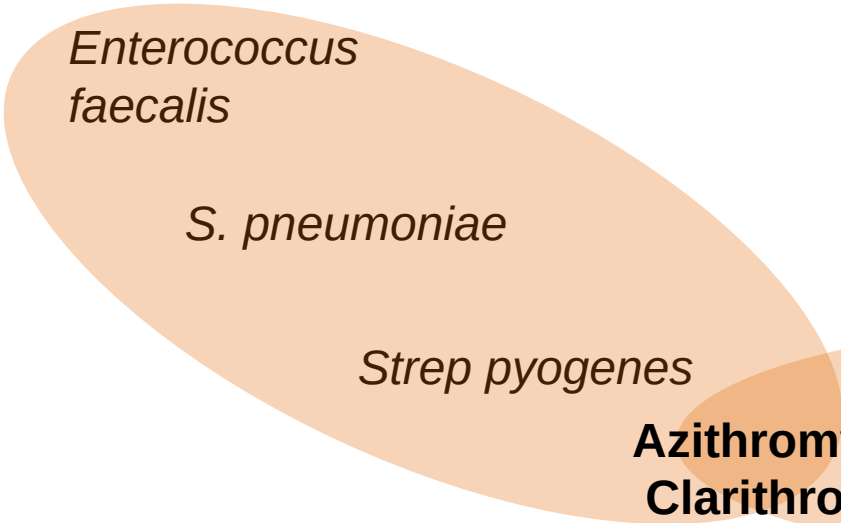
Enterobacter spp

Pseudomonas

Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)



*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Serratia

*Klebsiella
pneumoniae*

E. coli

*Proteus
mirabilis*

Haemophilus influenzae

Enterobacter spp

*Bacteroides
fragilis*

Pseudomonas

**Azithromycin &
Clarithromycin**

Gram-positive

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Gram-negative

Serratia

*Klebsiella
pneumoniae*

E. coli

*Proteus
mirabilis*

Haemophilus influenzae

Enterobacter spp

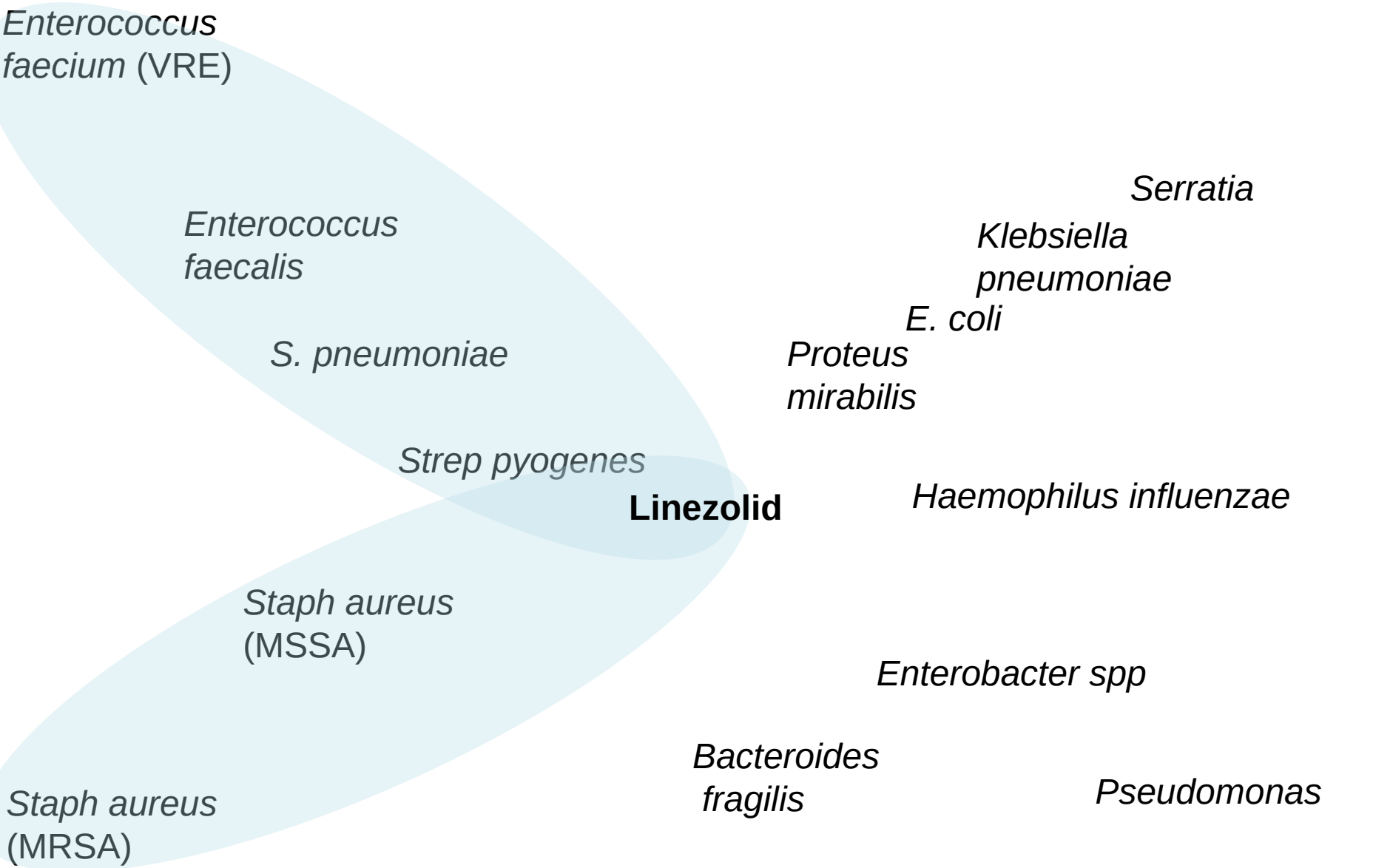
*Bacteroides
fragilis*

Pseudomonas

Clindamycin

Gram-positive

Gram-negative



Gram-positive

Enterococcus faecium (VRE)

Enterococcus faecalis

S. pneumoniae

Strep pyogenes

Staph aureus (MSSA)

Staph aureus (MRSA)

Gram-negative

Serratia

Klebsiella pneumoniae

E. coli

Proteus mirabilis

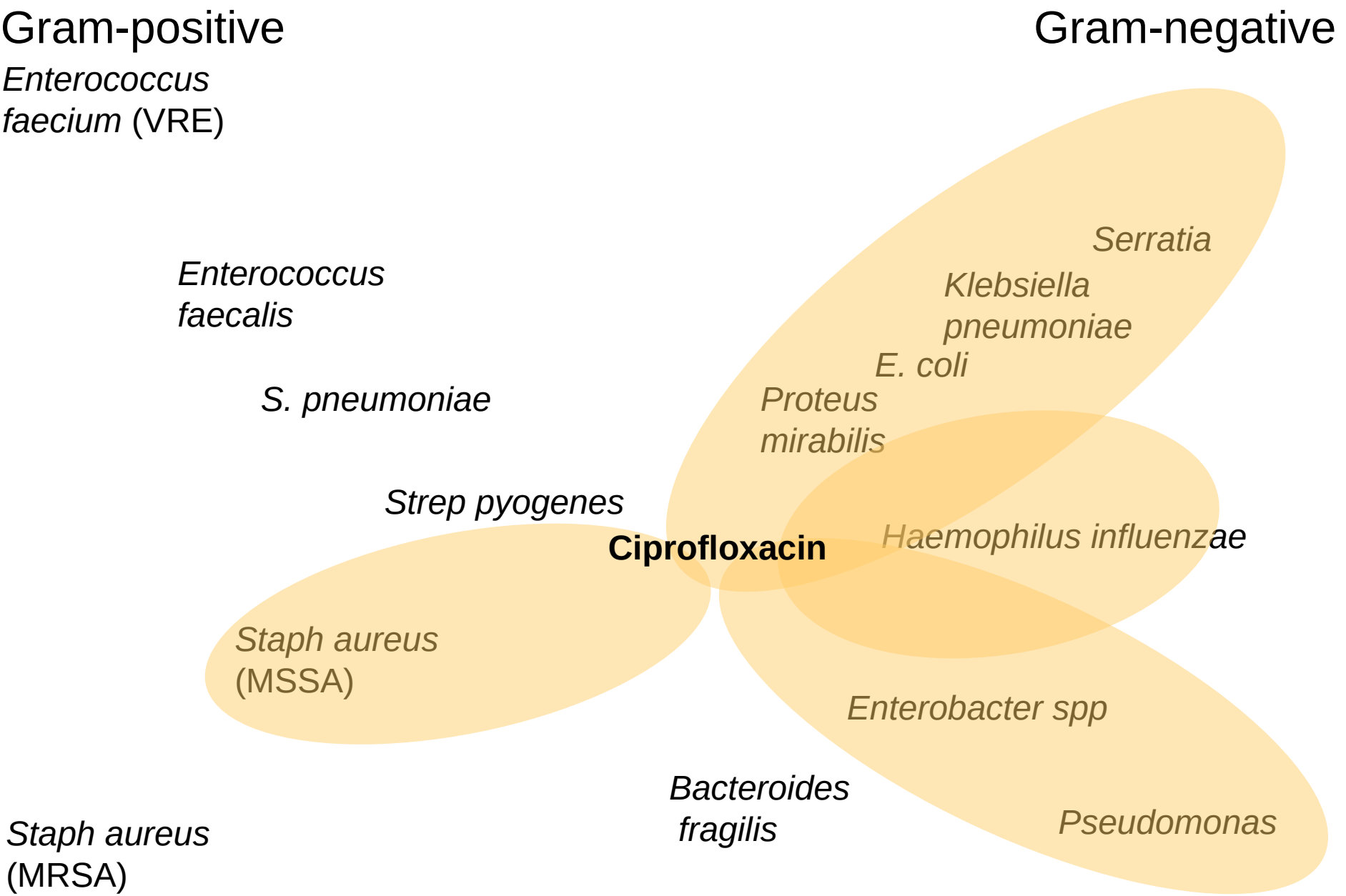
Haemophilus influenzae

Enterobacter spp

Bacteroides fragilis

Pseudomonas

Levofloxacin



Gram-positive

Enterococcus faecium (VRE)

Enterococcus faecalis

S. pneumoniae

Strep pyogenes

Staph aureus (MSSA)

Staph aureus (MRSA)

Gram-negative

Serratia

Klebsiella pneumoniae

E. coli

Proteus mirabilis

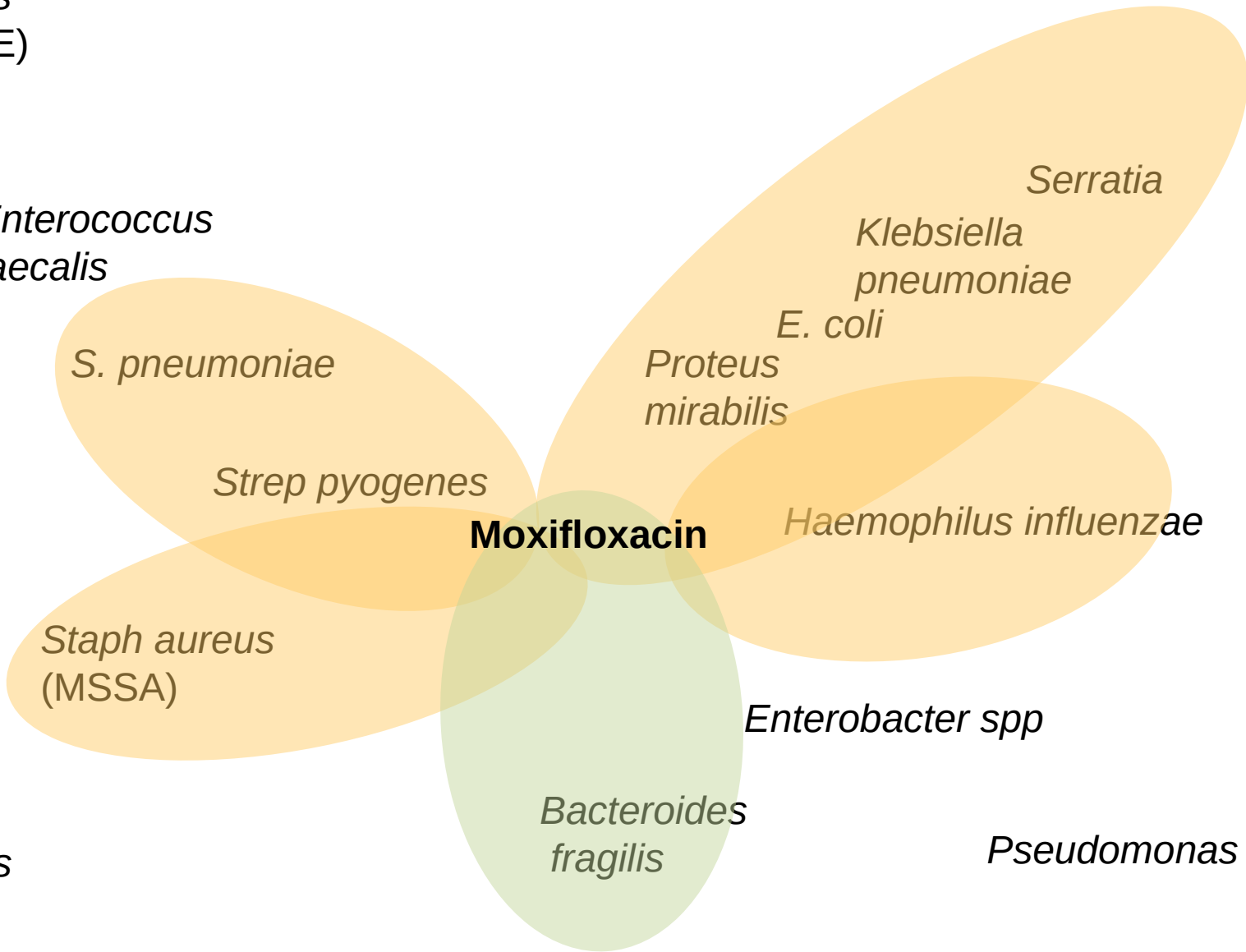
Haemophilus influenzae

Enterobacter spp

Pseudomonas

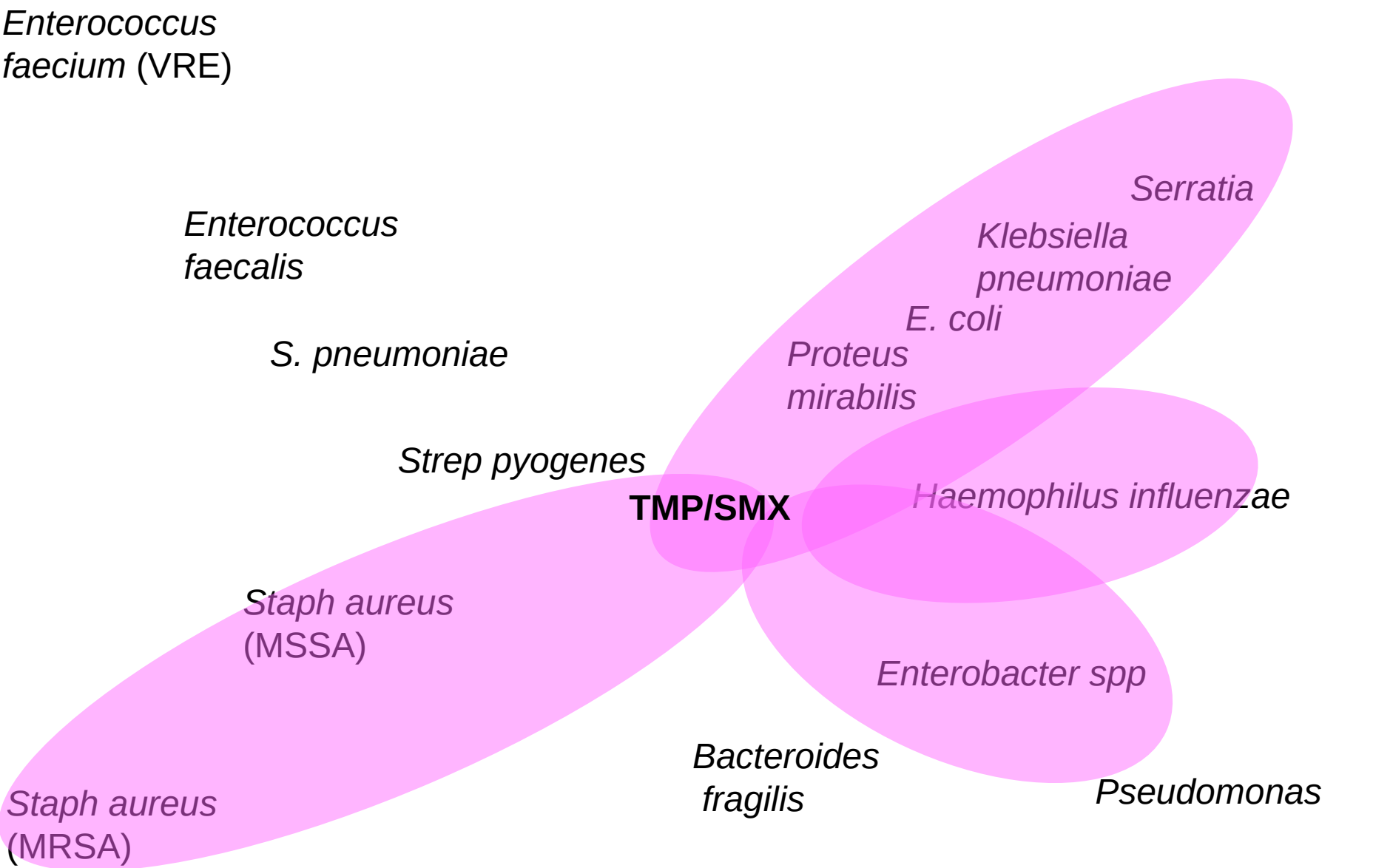
Moxifloxacin

Bacteroides fragilis



Gram-positive

Gram-negative



Gram-positive

Gram-negative

*Enterococcus
faecium* (VRE)

*Enterococcus
faecalis*

S. pneumoniae

Strep pyogenes

Staph aureus
(MSSA)

Staph aureus
(MRSA)

Serratia

*Klebsiella
pneumoniae*

E. coli

*Proteus
mirabilis*

Haemophilus influenzae

Enterobacter spp

Pseudomonas

Metronidazole

*Bacteroides
fragilis*

